Three Operations from T^4 to Observable Space

Demarcation from Discrete Emergence Models, with Case Study HLV

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Abstract

This document consolidates the demarcation of FFGFT against discrete emergence models (here: HLV by Krüger, August 2026) and clarifies which operations mediate between T^4 and the observable space $\mathbb{R}^3 \times \mathbb{R}_t$.

Central claim: The failure of discrete lattice models (class-X4 overlaps, missing operator specificity, open continuum limit) is not a surprise, but a structural consequence of treating geometric auxiliary constructs as ontological building blocks. FFGFT avoids these failures not by choosing a different lattice, but because the fundamental level T^4 is compact and geometrically complete — and the three operations leading from it to the observable space have strictly different logical status.

The three operations (Doc. 270 **[K]**):

1. **Type I** — Duality decompactification: $S^1_m \to \mathbb{R}_t$ via $\tilde{T} \cdot m = 1$. Information-preserving (embedding, not projection).

2. **Type II** — Geometric projection: $D3 \to D2 \to D1$. Lossy, irreversible. The step $D3 \to D2$ is identified as holographic projection.
3. **Type III** — Representational translation: $T^4 \leftrightarrow \mathscr{H}$ (Hilbert space/matrix). Bijective, lossless — upward movement, no geometric dimension reduction.

.1 Starting point: the HLV audit as case study

Marcel Krüger subjected his own Helix-Light-Vortex model (HLV) to four prospectively frozen mathematical tests in *Local-to-Global Compatibility on Aperiodic Cut-and-Project Carriers* (v1.4, August 2026). All key claims fail or remain open.

1. **Local-to-global compatibility:** 5,455 genuine volume overlaps (class X4) among 3,384 projected tetrahedra. Six witnesses certified with 120-digit arithmetic. *Failure.* **[E]**
2. **Operator specificity:** An ordinary periodic cubic lattice (CUBIC) outperforms HLV in spectral multiplet score (GAE-001A). *Rejected.* **[E]**
3. **Geometry-to-constitutive law:** Pure carrier geometry does not determine physical densities, masses, or couplings (Proposition 2.2). *Trivially negative.* **[E]**
4. **Finite-to-continuum:** A bounded graph has bounded spectrum, no UV limit. Mosco/resolvent convergence unproven. *Open.* **[E]**

Corollary (Krüger 4.5) [E]

Local validity of an operator does not imply global embeddability of the underlying carrier.

The structural cause

HLV attempts to use geometric bodies (3D tetrahedra from $\mathbb{R}^6$ projection) as ontological building blocks of physical space. It thereby inherits all four problems. The remedy is not a *better* lattice — it is a fundamental carrier that is geometrically complete and closed from the outset.

.2 The FFGFT fundamental level: T^4 with D4 lattice

Structure of T^4

The fundamental level of FFGFT is the compact 4-torus

$$T^4 = S^1_x \times S^1_y \times S^1_z \times S^1_m$$

with three spatial circles $S^1_{x,y,z}$ (radius R_H, quasi-continuous) and one mass circle S^1_m (radius $R_m \sim \xi \cdot \ell_P$, discrete). The fourth direction S^1_m is not a mass dimension alongside a separate time, but the substitute-time direction itself. Mass and time are two readings of *the same* dimension; $\tilde{T} \cdot m = 1$ is the translation dictionary (Doc. 270 **[K]**, Doc. 314 **[K]**). **[POSTULATE]**

The lattice on T^4 is the D4 lattice (Doc. 314 **[K]**): kissing number 24, $|\mathrm{Aut}(D_4)| = 1152$, triality as built-in $\mathbb{Z}_3$ symmetry (16 order-3 elements generate the $T^4/\mathbb{Z}_3$ orbifold with 9 fixed points). Negative result with theorem character: $5 \nmid 1152$, so $\mathrm{Aut}(D_4)$ contains no order-5 element **[K]**.

Why T^4 is flat — intrinsic vs. extrinsic curvature [K]

A torus carries *extrinsic* curvature when embedded in a higher-dimensional ambient space. The *intrinsic* Gaussian curvature, however, sums to zero pointwise — positive curvature (outer side) and negative curvature (inner side) cancel. By the Theorema Egregium, intrinsic curvature is an isometric invariant, independent of embedding.

Intrinsically flat torus [K]

T^4 is intrinsically flat: an interior observer measures no curvature. Angle sums are π, parallel geodesics remain parallel. T^4 is locally isometric to $\mathbb{R}^4$ — with global boundary identification, but without local curvature corrections.

Two direct consequences for FFGFT:

(a) D4 lattice through topology, not curvature. What the D4 lattice carries into the Hilbert space is topological data: winding numbers, mode content, triality. The distinction $\mathbb{Z}^4/D_4$ is spectral, not geometric (Doc. 314 **[K]**: "The torus is flat anyway — curvature was never the difference").

(b) Fourier analysis without curvature corrections. On the flat torus, Fourier modes $e^{ik\cdot x}$ form the natural basis; Laplacian eigenvalues are $|k|^2$ without curvature correction. This is the basis for the bijective, lossless Type III translation $T^4 \leftrightarrow L^2(T^4) \otimes \mathbb{C}^3$: Fourier decomposition on a flat torus is a complete isometric map.

This structurally distinguishes FFGFT from approaches like HLV, which embed geometric bodies into $\mathbb{R}^3$ and thereby generate overlap problems: the embedding question does not arise in FFGFT, because T^4 as an intrinsically flat, self-contained carrier requires no embedding into an outer space.

How information is encoded in the D4 lattice [K]

The geometry does not disappear when moving to the Hilbert space — it migrates into spectrum and fibre (Doc. 314 **[K]**):

Spectrum = theta series. The degeneracies of the Laplacian on flat T^4 are exactly the theta series of the D4 lattice: $\theta_{D_4}(q) = 1 + 24q^2 + 24q^4 + \ldots$ The kissing number 24 becomes the degeneracy of the first excitation $|k|^2 = 2$. Two flat tori ($\mathbb{Z}^4$ vs. D_4) are distinguishable by spectrum alone, without curvature **[K]**.

Fibre = triality. Triality is built in, not appended **[K]**: $|\mathrm{Aut}(D_4)| = 1152$, quotient to $W(D_4)$ is $6 = |S_3|$. The 16 order-3 elements generate the $\mathbb{Z}_3$ orbifold and act on every mode orbit as a $\mathbb{C}^3$ circulant. The fibre $\mathbb{C}^3$ follows from the lattice automorphism structure, not an additional postulate.

Winding numbers = topology. The reciprocity relation $\lambda_4 \cdot m = 2\pi$ holds mode-by-mode exactly **[K]**: winding number = mass number. Winding numbers are topological — invariant under deformation; fractal corrections accumulate only when unrolled, not when rolled up.

In one sentence (Doc. 314): "In the Hilbert space the lattice becomes flat, but not without consequence — what geometry carried, spectrum and fibre carry, and the division of labour between them is exactly determinable."

Where $\tilde{T} \cdot m = 1$ appears in the Hilbert space [K]

The time-mass duality appears in $\mathscr{H} = L^2(T^4) \otimes \mathbb{C}^3$ not as time evolution, but in three complementary forms:

(a) As the fourth winding number n_4. The multi-index $n = (n_1, n_2, n_3, n_4) \in \mathbb{Z}^4$ counts windings in all four torus directions **[B]** (Doc. 322). The mass dimension S^1_m carries the winding number n_4. The reciprocity relation $\lambda_4 \cdot m = 2\pi$ holds mode-by-mode exactly **[K]** (Doc. 314): winding number = mass number. $\tilde{T} \cdot m = 1$ is thus the Kaluza-Klein relation of the fourth torus direction, readable as n_4 quantum number in the spectrum.

(b) As the mass operator eigenvalue equation.

$$\hat{M}\,|n_\theta, n_\phi, p\rangle = m_i\,|n_\theta, n_\phi, p\rangle \quad \textbf{[K]} \tag{.1}$$

The eigenvalues are the particle masses (Doc. 322, Doc. 320). The duality appears as the spectrum of the operator.

(c) As radius-class separation in the spectrum. The splitting at $R_4 \neq R_{123}$ makes $\tilde{T} \cdot m = 1$ directly readable **[K]** (Doc. 314): the mass dimension generates a large mode spacing $\sim 1/R_m$, the spatial directions a quasi-continuous spacing $\sim 1/R_H$.

What does *not* appear in the Hilbert space

Time $\mathbb{R}_t$ itself. It is the result of the Type-I decompactification $S^1_m \to \mathbb{R}_t$ — a geometric downward operation *before* the Type-III translation. In the Hilbert space $L^2(T^4)$, only the torus lives; time appears only when leaving the Hilbert space and embedding the spectrum into $\mathbb{R}_t$. $\tilde{T} \cdot m = 1$ resides in the Hilbert space as n_4 winding number and mass operator eigenvalue structure, not as time evolution.

Where the fractal property resides in the Hilbert space [K] / [B]

The fractal property of FFGFT appears in the Hilbert space at three precisely separable locations:

(a) In the operator $\hat{F}$ itself — as self-similarity structure [B].

$$\hat{F} = \bigoplus_{n=1}^{N} S_n \circ L_n, \qquad S_n = r_n U_n, \qquad r_n = \xi^n \in (0,1) \tag{.2}$$

The scale operators S_n implement discrete scale transformations; each level n is a contraction of the previous by r_n — this is self-similarity as an operator. $\hat{F}$ encodes the sub-Planck geometry as a hierarchical convolution of scale levels down to the minimum length scale $L_0 = \xi \cdot \ell_P$ (Doc. 180, 327 **[K]**).

(b) In the fractal dimension $D_f = 3 - \xi$ — as spectral property [K].

$$D_f = \inf\left\{s \,:\, \sum_i \lambda_i^s < \infty\right\} = 3 - \xi \approx 2.9999 \quad \textbf{[K]} \tag{.3}$$

D_f is defined via the Hausdorff measure of the spectrum of $\hat{F}$ (Doc. 322): the spectrum on $L^2(T^4)$ has a fractal fine structure. The value is spatial ($3 - \xi$, three dimensions), not four-dimensional. Local action: $D_f^{\text{space}} = 3 - \xi$ with 6.7×10^{-5} relative (Doc. 314 **[K]**).

(c) In the interface factor K_{frac} — as scale bridge [K].

$$K_{\text{frac}} = 1 - 100\xi = \frac{74}{75} \approx 0.9867 \tag{.4}$$

K_{frac} is cumulative (Doc. 133): it accumulates over the RG running across 17 orders of magnitude and acts at the transition from geometric (bare) to measured (SI) mass: $m_i^{\text{SI}} = K_{\text{frac}} \cdot C_{\text{conv}} \cdot m_i^{\text{bare}}$.

Decisive separation [K]

In the *rolled-up* T^4 (Hilbert space) there are **no** fractal corrections — winding numbers are purely topological (Doc. 314 **[K]**). The fractal property appears only when unrolling (Type I: $S_m^1 \to \mathbb{R}_t$) as an accumulated defect. In the Hilbert space itself, $\hat{F}$ is a bounded self-adjoint operator with real spectrum **[B]** — the fractality resides in the scale hierarchy of the operator structure ($r_n = \xi^n$), not in the carrier $L^2(T^4)$.

.3 No embedding problem in T^4

T^4 is compact, complete, and geometrically self-contained. There is no embedding problem because T^4 does not need to be *embedded* in an outer space: it *is* the fundamental level. The class-X4 errors of HLV arise because projection elements must be embedded into an external $\mathbb{R}^3$. In FFGFT this question does not arise.

Type III: representational translation upward

The translation $T^4 \leftrightarrow \mathscr{H} = L^2(T^4) \otimes \mathbb{C}^3$ (Doc. 230, 314, 322) is a bijective, lossless upward movement (Type III, Doc. 270 **[K]**). It adds no geometric extra dimensions, but expresses the same T^4 content in a higher-dimensional language. The spectrum of $\hat{F}$ (Doc. 322, self-adjoint **[B]**, Doc. 327) has the particle masses as eigenvalues; $\xi = \lambda_{\min}(\hat{F}_{D_4})$ is the smallest eigenvalue **[B]**.

.4 The three operations from T^4 to the observable space

Type I: duality decompactification

$$S^1_m \xrightarrow{\tilde{T}\cdot m=1} \mathbb{R}_t$$

The unrolling of the compact mass dimension into the non-compact time direction is an *embedding*, not a projection. No mode content is lost; time arises as an additional non-compact direction **[K]**.

Type II: geometric projection (lossy)

The further steps $D3 \to D2 \to D1$ are genuine geometric projections of spatial directions: lossy, irreversible. The step $D3 \to D2$ is identified as holographic projection (Doc. 270 **[K]**). It is not fundamental.

Classification of the holographic step

The step $D3 \to D2$ is *not* the fundamental level of FFGFT. It is a lossy observer projection — physically relevant for surface phenomena (black hole membrane, Doc. 325), but not for the derivation of masses and couplings anchored on T^4. The two-dimensional projection surface is not a fundamental object.

Three-way comparison

Type	Operation	Direction	Status
I	Duality decompactification $S^1_m \to \mathbb{R}_t$	geometrically down	information-preserving **[K]**
II	Geometric projection $D3 \to D2 \to D1$	geometrically down	lossy **[K]**
III	Representational translation $T^4 \leftrightarrow \mathscr{H}$	representationally up	bijective, lossless **[K]**

.5 Open questions in the flat Hilbert space — status

Question 1: Haar measure and D4 lattice [B]

The canonical Haar measure on T^4 is compatible with the D4 spectral structure (scripts 330-1, all 7/7): Fourier modes are exactly orthogonal; D4 contains only even norm shells; Aut$(D_4) \subset$ Iso(T^4) leaves the Haar measure invariant. The $\mathbb{Z}_3$ orbifold breaks translation invariance and splits the 24-fold degeneracy to $9 \oplus 8 \oplus 4 \oplus 2 \oplus 1$ (Doc. 314 **[K]**) — but only as fine structure $\sim \xi \approx 10^{-4}$, below any measurement threshold. **[B]**

Question 2: Orbifold fixed points [K] / [S]

Nine fixed points of $T^4/\mathbb{Z}_3$ follow from the Lefschetz fixed-point formula (script 330-2, B1): $|\det(1-A)| = |(1-\omega)(1-\omega^2)|^2 = 9$. Neutrino localisation: $\nu_1 \leftrightarrow F_2$, $\nu_2 \leftrightarrow F_5$, $\nu_3 \leftrightarrow F_7$ **[K]** (Doc. 322). What remains **[S]**: back-reaction of Hawking emission on the fixed-point structure during $M \to M_{\text{coll}} = m_{\text{P}}/\sqrt{2}$ (explicitly open, Doc. 325).

Question 3: Fractal weighting r_n and $L^2(T^4)$ [B]

$r_n = \xi^n$ is position-independent: $\hat{F}$ does *not* break the translation invariance of $L^2(T^4)$ — only the $\mathbb{Z}_3$ orbifold does (and only as fine structure $\sim \xi$). $\hat{F}$ is bounded ($\|\hat{F}\| \leq \sum r_n < \infty$) and self-adjoint on $\mathscr{H}$ **[B]** (Doc. 327). **[B]**

Status of the three questions

Question 1 (Haar/D4): **[B]** — fully resolved.
Question 2 (fixed points): **[K]** for 9 fixed points and ν-localisation; **[S]** for back-reaction during evaporation (explicitly open, Doc. 325).
Question 3 ($r_n/\hat{F}$): **[B]** — $\hat{F}$ bounded and self-adjoint; r_n does not break translation invariance.

.6 Why Krüger's four problems do not arise in FFGFT

1. **Class-X4 overlaps:** arise because HLV embeds projection elements into $\mathbb{R}^3$. In FFGFT, T^4 is the fundamental level; no embedding into an outer space occurs.
2. **Missing operator specificity:** In FFGFT the spectrum is determined by the D4 lattice with built-in triality. The negative result $5 \nmid |\text{Aut}(D_4)|$ is itself a specific theorem about the spectrum **[K]**.
3. **Geometry without physics:** T^4 alone also does not fix coupling constants. The physics comes from the combination T^4-spectrum + $\xi = \lambda_{\text{min}}(\hat{F}_{D_4})$.
4. **Continuum limit:** No limit problem, because Type I is an embedding (no limit passage needed) and Type III is bijective (no information loss). Only Type II is lossy — but this is known and documented (Doc. 270).

.7 What remains from the comparative discussion

Structurally secured: The four HLV problems do not arise in FFGFT — due to T^4 as intrinsically flat, self-contained carrier, bijective Type-III translation, and Type-I embedding $S^1_m \hookrightarrow \mathbb{R}_t$.

Correct from the comparative discussion: The holographic projection $D3 \to D2$ (Type II) is lossy and not fundamental; mass and time are not encoded in the 2D surface but arise through Type I ($\tilde{T} \cdot m = 1$) and the recursion ξ^n.

Correction vs. uncritical adoption: There is no "two-dimensional projection surface as fundamental object" in FFGFT. The corpus has a clear ontological hierarchy: T^4 as primary reality (Doc. 152 **[POSTULATE]**), not a 2D surface. $\tilde{T} \cdot m = 1$

is the Kaluza-Klein relation of the mass dimension, not a "reading principle on a screen".

Verification scripts: `pruef_330_1_haar_d4.py` (7/7), `pruef_330_2_fixpunkte_rn.py` (9/9).

Status overview

Element	Status
$T^4 = S^1_x \times S^1_y \times S^1_z \times S^1_m$ as fundamental level	**[POSTULATE]**
D4 lattice on T^4; kissing number 24; $\lvert\mathrm{Aut}(D_4)\rvert = 1152$	**[K]**
T^4 intrinsically flat; spectral structure from topology	**[K]**
Information in spectrum (theta series) and fibre (triality)	**[K]**
$\tilde{T} \cdot m = 1$ as n_4 winding number and mass operator eigenvalue	**[K]**
Fractal property in $\hat{F}$ structure ($r_n = \xi^n$), not in $L^2(T^4)$	**[B]**
$D_f = 3 - \xi$; no fractal corrections in rolled-up T^4	**[K]**
Three operation types (Doc. 270): Type I/II/III	**[K]**
Type III bijective ($T^4 \leftrightarrow \mathscr{H}$, Doc. 322/314)	**[K]**
HLV class-X4 errors structurally excluded (no embedding problem)	**[B]**
Continuum limit structurally resolved by Type I (embedding)	**[K]**